Day: M T W T F S Date: __/__/20__

(10) **a** T_1 = Shear in Y-direction with factor of $\frac{1}{4}$ T_2 = Shear in Y-direction with factor of $\frac{3}{5}$

$$\left(\frac{1}{4}\right) T_1 = \begin{bmatrix} 1 & 0 \\ 1/4 & 1 \end{bmatrix} \quad \left(\frac{3}{5}\right) T_2 = \begin{bmatrix} 1 & 0 \\ 3/5 & 1 \end{bmatrix}$$

standard matrices of T_1 & T_2

$$[T_1 \circ T_2] = [T_1][T_2] = \begin{bmatrix} 1 & 0 \\ 17/20 & 1 \end{bmatrix}$$

$$[T_2 \circ T_1] = [T_2][T_1] = \begin{bmatrix} 1 & 0 \\ 17/20 & 1 \end{bmatrix}$$

$\therefore T_1 \circ T_2 = T_2 \circ T_1$, the operators commute

b T_1 = Shear in y-direction with factor of 1/4 T_2 = Shear in x-direction with factor of 3/5.

$$\left(\frac{1}{4}\right) T_1 = \begin{bmatrix} 1 & 0 \\ 1/4 & 1 \end{bmatrix} \quad \left(\frac{3}{5}\right) T_2 = \begin{bmatrix} 1 & 3/5 \\ 0 & 1 \end{bmatrix}$$

$$[T_1 \circ T_2] = [T_1][T_2] = \begin{bmatrix} 23/20 & 3/5 \\ 1/4 & 1 \end{bmatrix}$$

$$[T_2 \circ T_1] = [T_2][T_1] = \begin{bmatrix} 1 & 3/5 \\ 1/4 & 23/20 \end{bmatrix}$$

$\therefore T_1 \circ T_2 \neq T_2 \circ T_1$, the operata's don't commute.